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Supplementary information

The evolution of private reputations in information-abundant landscapes

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Supplementary Information for

The evolution of private reputations in abundant information landscapes

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1 Full model description

1.1 Strategy dynamics

Interactions. We consider a well-mixed population of individuals engaged in pairwise, one-shot interactions that take the form of donation games, also known as simplified Prisoner's Dilemma. In each interaction, one individual is the potential donor of a cooperative act, and the other is the recipient. A donor can either cooperate (C), providing a benefit b to the recipient at their own cost c ($0 < c < b$), or defect (D), incurring no cost and providing no benefit to the recipient.

Interaction strategies. A donor can behave either unconditionally, choosing an action without considering the recipient's reputation, or conditionally, choosing an action based on the recipient's reputation. We model the reputations of individuals as binary values 0 or 1, which we refer to as bad (B) or good (G), respectively. We describe the interaction strategies by the tuple $s = (s^B, s^G)$, where s^R represents the decision to donate ($s^R = 1$) or not ($s^R = 0$) to a recipient with reputation $R \in \{B, G\}$. Throughout the paper, we consider the following strategies:

- Cooperators (denoted ALLC or x): Individuals who always cooperate; strategy
$$s_x = (1, 1).$$
- Defectors (denoted ALLD or y): Individuals who always defect; strategy
$$s_y = (0, 0).$$
- Discriminators (denoted $\text{DISC}_{q,M}$ or z): Individuals who cooperate only with those whom they view as good; strategy
$$s_z = (0, 1).$$
The parameters q and M in $\text{DISC}_{q,M}$ specify the reputation aggregation rule described below (see Aggregation of multiple assessments).
- Probabilistic cooperators (denoted probC_p or w): Individuals who cooperate with some fixed probability p regardless of the opponent; strategy
$$s_w = (p, p).$$

Replicator dynamics. We assume that each individual follows one of the above strategies and that individuals update their strategies by payoff-biased imitation. We describe the dynamics of strategy competition in the limit of an infinitely large population using replicator equations[1]. Let f_i represent the frequency of strategy i. The dynamics of f_i are described by the ODE

$$\frac{df_i}{dt} = f_i \left(\Pi_i - \sum_{s \in S} f_s \Pi_s \right), \tag{S1}$$

where S is the set of all strategies considered in a population and Π_i is the fitness of strategy i.

Fitness. As in prior models[2], we compute the fitness of an individual of strategy i as the average payoff that the individual gains from an interaction, given by

$$\Pi_i = \sum_{j \in S} f_j (b a_{ji} - c a_{ij}), \tag{S2}$$

where ba_{ji} is the benefit received by a strategy-i player from a strategy-j player, and ca_{ij} is the cost incurred by an i player to cooperate with a j player. The term a_{ij} denotes the expected action of an individual using strategy i toward an individual using strategy j.

Actions. The expected action, defined as the probability that an individual with strategy i cooperates with someone using strategy j, is given by

$$a_{ij} = s_i^B(1 - r_{ij}^*) + s_i^G r_{ij}^*, \tag{S3}$$

where r_{ij}^* denotes the probability that i observes j as good. Specifically, r_{ij}^* is the expected reputation of strategy j in the eyes of strategy i at the equilibrium of the reputation dynamics described below. In other words, r_{ij}^* is the proportion of individuals of strategy i that have a positive opinion of a random individual of strategy j. Thus, while r_{ij} is the binary reputation of a single individual and only takes the value of 0 or 1, the expected reputation of a strategy r_{ij}^* can take values between 0 and 1.

The expected action of the four strategies we considered can be computed by substituting s_i into Eq. S3:

$$\text{ALLC:} \quad a_{x,j} = 1 \tag{S4a}$$

$$\text{ALLD:} \quad a_{y,j} = 0 \tag{S4b}$$

$$\text{DISC}_{q,M}\text{:} \quad a_{z,j} = r_{zj}^* \tag{S4c}$$

$$\text{probC}_p\text{:} \quad a_{w,j} = p \tag{S4d}$$

Note that the actions of unconditional strategies (x, y, and w) are independent of the opponent (j), while the expected action of discriminators (z) depends on how they perceive others. Specifically, the term r_{zj}^* represents the probability that a discriminator has a good perception of a randomly selected individual with strategy j.

1.2 Reputation dynamics

Monitoring reputations. After a round of pairwise games—that is, once everyone has played a game with everyone else, once as a donor and once as a recipient—individuals privately update their views of every member of the population. To assign reputations, individuals *observe* the actions of others and assess each action as good or bad according to a *social norm*.

Social norms. A *social norm* is an assessment rule used to assign reputations. A 'first-order' social norm maps the action of the observed donor to a moral judgment; in contrast, a 'second-order' social norm maps the action of the donor and the reputation of the recipient to a moral judgment. We focus on one first-order norm and three second-order norms, as described below, although our model could be easily expanded to include higher-order social norms[3].

We describe each social norm by a vector

$$d = \{d_{CG}, d_{CB}, d_{DG}, d_{DB}\} \in \{0,1\}^4,$$

where the entry d_{AR} denotes whether an action $A \in \{C,D\}$ towards someone with reputation $R \in \{B,G\}$ is considered bad or good. For example, $d_{CG} = 1$ means that cooperating with a good recipient is perceived as good, and $d_{CB} = 0$ means that cooperating with a bad recipient is considered bad.

Previous studies have shown that the social norms that foster cooperation and are evolutionarily stable share the principle of rewarding cooperation and punishing defection towards good individuals[4] (i.e., they are all of the form $d = \{1,*,0,*\}$, where $*$ can be either 0 or 1). We explore the social norms that follow this principle but vary in their assessment of actions towards individuals with a bad reputation:

- Scoring[5,6], which rewards cooperation regardless of the recipient's reputation (hence a first-order norm):

$$d = \{1,1,0,0\}\ .$$

- Stern-Judging[7–9], which rewards defection against bad recipients:

$$d = \{1,0,0,1\} .$$

- Simple-Standing[10,11], which rewards any action against bad recipients:

$$d = \{1,1,0,1\} .$$

- Shunning[12], which punishes any action against bad recipients:

$$d = \{1,0,0,0\} .$$

Observations. In every round of observations, each observer independently observes M interactions of every donor to update their view of the donor. Specifically, each observer samples M interactions at random and with replacement from the set of all interactions of the donor in the previous round. As we describe below, the observer assesses each of the M actions by applying the social norm to each interaction independently and then aggregates the assessments across the M observations to assign the donor a new reputation.

Assessment of a single observation. An observation made by an observer with strategy i of an interaction between a donor with strategy j and a recipient with strategy k can only be one of the following cases:

Action of a strategy-j donor toward a strategy-k recipient	Reputation of a strategy-k recipient according to a strategy-i observer	Probability	Relevant entry of the social norm
C	G	$a_{jk}\, r_{ik}$	d_{CG}
C	B	$a_{jk}\, (1-r_{ik})$	d_{CB}
D	G	$(1-a_{jk})\, r_{ik}$	d_{DG}
D	B	$(1-a_{jk})\, (1-r_{ik})$	d_{DB}

Table S1: Possible observations made by an observer i of a donor j interacting with a recipient k. The first column describes the donor's action: cooperate (C) or defect (D). The second column describes the recipient's reputation: good (G) or bad (B). The third column describes the probability that such action and reputation are observed. The fourth column shows the corresponding entry of the social norm.

Thus, the probability p_{ijk} that an observer with strategy i assesses as good a randomly selected interaction between a donor with strategy j and a recipient with strategy k is given by

$$p_{ijk} = a_{jk}(r_{ik}d_{CG} + (1-r_{ik})d_{CB}) + (1-a_{jk})(r_{ik}d_{DG} + (1-r_{ik})d_{DB}), \tag{S5}$$

where d_{AR} is prescribed by the social norm, a_{jk} is the action of the donor towards the recipient, and r_{ik} is the reputation of the recipient in the eyes of the observer, as described in Table S1.

Under different social norms (see Social norms), Eq. S5 takes diverse functional forms:

$$\text{Scoring:} \quad p_{ijk} = a_{jk} , \tag{S6a}$$

$$\text{Stern-Judging:} \quad p_{ijk} = a_{jk}r_{ik} + (1-a_{jk})(1-r_{ik}) , \tag{S6b}$$

$$\text{Simple-Standing:} \quad p_{ijk} = a_{jk} + (1-a_{jk})(1-r_{ik}) , \tag{S6c}$$

$$\text{Shunning:} \quad p_{ijk} = a_{jk}r_{ik} . \tag{S6d}$$

Assessment of a single observation between discriminators. Here we consider the special case in which both the donor and the observer are discriminators (i.e., $i = j = z$). By substituting Eq. S4c into Eqs. S6, we obtain

$$\text{Scoring:} \quad p_{zzk} = r_{zk} \,, \tag{S7a}$$

$$\text{Stern-Judging:} \quad p_{zzk} = r_{zk}^2 + (1 - r_{zk})^2 \,, \tag{S7b}$$

$$\text{Simple-Standing:} \quad p_{zzk} = r_{zk} + (1 - r_{zk})^2 \,, \tag{S7c}$$

$$\text{Shunning:} \quad p_{zzk} = r_{zk}^2 \,. \tag{S7d}$$

Eqs. S7 describe the probability that a discriminator assesses as good an interaction between a discriminator and a random recipient of strategy k. In other words, Eqs. S7 describe the classical private reputation dynamics between discriminators using a single observation[13,14]. A donor earns a good reputation in the eyes of an observer after a single observation if the following occurs:

- Under Scoring, a donor earns a good reputation if the donor interacts with a good individual (which occurs with probability r_{zk}) and, thus, cooperates.
- Under Stern-Judging, a donor earns a good reputation if the donor interacts with an individual who has a good reputation in the eyes of both the observer and the donor (which occurs with probability r_{zk}^2) and the donor cooperates, or if the donor interacts with an individual who has a bad reputation in the eyes of both (which occurs with probability $(1 - r_{zk})^2$) and the donor defects.
- Under Simple-Standing, a donor earns a good reputation if the donor interacts with an individual who has a good reputation in the eyes of the donor (which occurs with probability r_{zk}) and cooperates, or if the donor interacts with an individual who has a bad reputation in the eyes of both the observer and the donor (which occurs with probability $(1 - r_{zk})^2$) and defects.
- Under Shunning, a donor earns a good reputation if the donor interacts with an individual who has a good reputation in the eyes of both the observer and the donor (which occurs with probability r_{zk}^2).

In other words, under Scoring, the only requirement to gain a good reputation is to cooperate and, for discriminators, this means to encounter a good recipient; while under Stern-Judging, the requirement to gain a good reputation is to coordinate the private views between the observer and the donor.

Aggregation of multiple assessments. To assign a binary reputation based on multiple observed actions, each of which is assessed as good or bad according to the observer's social norm, the observer must somehow aggregate information across observations. We study an *aggregation rule* defined by a number of observations M and a strictness threshold $q \in [0,1]$, which determines the minimum number of good actions ($\lceil qM \rceil$ out of the M observed) needed to confer a good reputation.

The probability that an observer with strategy i assesses as good an action of a donor with strategy j towards a randomly selected recipient is:

$$p_{ij} = \sum_{k \in S} f_k p_{ijk}. \tag{S8}$$

An observer assesses M actions of a donor and assigns the donor a good reputation if at least $q \cdot M$ assessments are good. We let (q_i, M_i) be the aggregation rule corresponding to strategy i; in other words, a strategy-i observer makes M_i observations and uses a tolerance threshold q_i. Given that the M_i observed interactions are chosen uniformly at random and with replacement, the probability ρ_{ij} that a strategy-i observer assigns a strategy-j donor a good reputation—which is equivalent to the probability that the observer assesses at least $q_i \cdot M_i$ actions of a strategy-j donor as good—is given by

$$\rho_{ij} = \sum_{m=\lceil q_i M_i \rceil}^{M_i} \binom{M_i}{m} p_{ij}^m (1 - p_{ij})^{M_i - m}. \tag{S9}$$

Errors. We allow for errors in strategy execution and assessment[15]. With probability ε, an intended cooperative act is executed as defection; in contrast, an intended defection is always executed correctly. To account for this execution error, we let

$$a_{ij}\big|_{\text{with error}} = (1-\varepsilon)a_{ij}\big|_{\text{without error}},$$

where $a_{ij}|_{\text{without error}}$ is defined in Eq. S3.

With probability α, the observer incorrectly evaluates a good action as bad or vice versa. To account for this assessment error, we let

$$\begin{aligned} p_{ij}\big|_{\text{with error}} &= (1-\alpha)\, p_{ij}\big|_{\text{without error}} + \alpha\left(1 - p_{ij}\big|_{\text{without error}}\right) \\ &= \alpha + (1-2\alpha) p_{ij}\big|_{\text{without error}}, \end{aligned}$$

where $p_{ij}|_{\text{without error}}$ is defined in Eq. S8. In other words, with probability $1-\alpha$, an observer uses the correct assessment (which is good with probability p_{ij}), and with probability α, they use the opposite assessment (which is good with probability $1-p_{ij}$).

Reputation updating. In this section, we derive the reputation dynamics stated in the main text (Eq. 7). To do so, we study a discrete-time process for a finite population of N individuals, and then we derive a continuum limit by re-scaling time as $N \to \infty$.

In the discrete-time, finite-population process, each individual $A \in \{1 \dots N\}$ has a view of every other individual $B \in \{1 \dots N\}$; population views are represented by a reputation matrix r_{AB}. A value of 1 corresponds to A having a good view of B; a value of 0 corresponds to A having a bad view of B. (Here, and elsewhere in this section, capital letters A and B are used for individual players, in contrast to lowercase letters i and j, which are reserved for strategies.)

Initially, we consider discrete time steps of size denoted $\Delta\tau$. In each time step, a single, solitary observation occurs: that is, a randomly chosen individual resolves to update their view of a random donor, by observing the donor's behavior toward a random recipient. Errors in both execution (with probability ε) and assessment (with probability α) may occur; errors are independent and randomly drawn for each observation. After obtaining the expected change in r_{AB} as a result of a single observation, we will take the limit as N approaches infinity and sum over all players to obtain a mean-field description of the chance that a randomly chosen individual with strategy i will have a positive view of a randomly chosen player with strategy j. Finally, we will take the limit as duration of time steps, $\Delta\tau$, approaches zero, resulting in the ordinary differential equation, Eq. 7, given in the main text.

We first focus on the classical case, $M = 1$. In a given time step, a single interaction occurs, in which some individual acts as a donor toward some recipient. With probability $1/N$, that donor is B; with probability $1/N$, that recipient is some particular $D \in \{1 \dots N\}$. (The particular identity of D does not matter; we will sum over all values of D.) An observer uses her view of D, and B's action toward D, to update her view of B. With probability $1/N$, that observer is A. Thus, the total probability that, in a given time step, A updates her view of B based on an interaction with D is $1/N^3$.

The expected value of A's new opinion of B's reputation will be p_{ABD}; as a result of reputation updating, r_{AB}, which is A's current view of B, is erased. To assess what happens to r_{AB} in a time step $\Delta\tau$, we average over all possible recipients D. Defining $p_{AB} = \sum_D p_{ABD}/N$ allows us to write

$$\begin{aligned} \mathbb{E}[r_{AB}(\tau+\Delta\tau)] &= r_{AB}(\tau) + \Delta\tau \sum_D \frac{1}{N^3}\left(p_{ABD} - r_{AB}(\tau)\right) \\ &= r_{AB}(\tau) + \Delta\tau \frac{1}{N^2}(p_{AB} - r_{AB}(\tau)). \end{aligned} \tag{S10}$$

This tallying can immediately be extended to our model in which A judges B not based on a single interaction of B with D, but rather based on B's interactions with M_A recipients $D_\mu, \mu \in \{1 \dots M_A\}$. In our mean-field approximation, this is mathematically equivalent to allowing B to play with some larger number of

individuals each round but allowing A to consider only M_A of those actions. Let p_{ABD_μ} be A's assessment of B's action toward some D_μ. To obtain the probability that A assigns B a good reputation, we first consider the probability that, when B interacts with randomly chosen recipients $D_1 \dots D_{M_A}$, exactly m of those interactions are assessed as "good". We sum this over all relevant values of m, then sum over all possible values each D_μ can take. We define

$$\rho_{AB} = \frac{1}{N^{M_A}} \sum_{D_1=1}^{N} \cdots \sum_{D_{M_A}=1}^{N} \sum_{m=\lceil q_A M_A \rceil}^{M_A} \left(\prod_{\mu=1}^{m} p_{ABD_\mu} \right) \left(\prod_{\mu=m+1}^{M_A} (1 - p_{ABD_\mu}) \right). \tag{S11}$$

Then Eq. S10 becomes:

$$\mathbb{E}[r_{AB}(\tau + \Delta\tau)] = r_{AB}(\tau) + \Delta\tau \frac{1}{N^2}(\rho_{AB} - r_{AB}(\tau)). \tag{S12}$$

Finally, the change in r_{ij}, the *average* standing of a strategy-j player in the eyes of a strategy-i player, can be obtained. Let i be player A's strategy and j be player B's strategy, and denote by σ_i the set of all i-players (likewise σ_j for j-players). Further, let $n_i = Nf_i$ be the number of i-players (likewise $n_j = Nf_j$ for j-players). Then

$$\begin{aligned} \mathbb{E}[r_{ij}(\tau + \Delta\tau)] &= \frac{1}{n_i n_j} \sum_{A \in \sigma_i} \sum_{B \in \sigma_j} \mathbb{E}[r_{AB}(\tau + \Delta\tau)] \\ &= r_{ij}(\tau) + \Delta\tau \frac{1}{N^2}(\rho_{ij} - r_{ij}(\tau)) \\ \therefore \mathbb{E}[r_{ij}(\tau + \Delta\tau) - r_{ij}(\tau)] &= \Delta\tau \frac{1}{N^2}(\rho_{ij} - r_{ij}(\tau)). \end{aligned} \tag{S13}$$

Taking the limit as $\Delta\tau \to 0$ yields

$$\frac{dr_{ij}}{d\tau} = \frac{1}{N^2}(\rho_{ij} - r_{ij}). \tag{S14}$$

Re-scaling time so that N^2 update events happen per time step, which corresponds to (on average) everyone updating their opinion of everyone else, allows us to write $t = \tau/N^2$, and so we obtain the reputation dynamics stated in the main text (Eq. 7):

$$\frac{dr_{ij}}{dt} = \rho_{ij} - r_{ij}. \tag{S15}$$

1.3 Norm dynamics

We now consider the coevolution of social norms and strategies, when both are updated independently by imitation. We characterize individuals by their 'type', consisting of a strategy and a social norm. We denote by f_{sn} the frequency of type (s,n), i.e., individuals using strategy s and social norm n. We assume that an imitation event happens after a focal individual compares its payoff with a role model. Then, such a focal individual imitates either the strategy of the role model with probability w, or the social norm of the role model with probability $1-w$.

During a strategy imitation event, the frequency f_{sn} can change as follows:

- f_{sn} increases when individuals of type (i,n) with strategy $i \neq s$ imitate the strategy of individuals of type (s,j), for any norm j;
- f_{sn} decreases when individuals of type (s,n) imitate the strategy of individuals of type (i,j) with strategy $i \neq s$, for any norm j.

During a norm imitation event, the frequency f_{sn} can change as follows:

- f_{sn} increases when individuals of type (s,j) with social norm $j \neq n$ imitate the norm of individuals of type (i,n), for any strategy i.
- f_{sn} decreases when individuals of type (s,n) imitate the norm of individuals of type (i,j) with norm $j \neq n$, for any strategy i.

Note that, because the imitation of strategies and social norms are independent events, individuals cannot change both components at once. For example, a type (i,j) with $i \neq s$ and $j \neq n$ can only convert to type (s,n) after two successful imitation events (i becoming s and j becoming n, in any order).

A type (i,j) successfully imitates one of the two components of type (s,n) with probability ϕ_{sn}^{ij}, which is governed by comparing the payoffs of the two types using the Fermi function:

$$\phi_{sn}^{ij} = \frac{1}{1+e^{\beta(\Pi_{ij}-\Pi_{sn})}},$$

where β represents the strength of selection[16]. The dynamics of the frequency of type f_{sn} are then described by the ODE:

$$\frac{df_{sn}}{dt} = \underbrace{w\sum_{i\neq s}\sum_{j}\left(\overbrace{f_{in}f_{sj}\phi_{sj}^{in}}^{i\to s} - \overbrace{f_{sn}f_{ij}\phi_{ij}^{sn}}^{s\to i}\right)}_{\text{strategy imitation}} + \underbrace{(1-w)\sum_{i}\sum_{j\neq n}\left(\overbrace{f_{in}f_{sj}\phi_{in}^{sj}}^{j\to n} - \overbrace{f_{sn}f_{ij}\phi_{ij}^{sn}}^{n\to j}\right)}_{\text{social norm imitation}} \tag{S16}$$

The first term of Eq. S16 describes the contribution of strategy imitation (w) to the change of $f_{s,n}$, while the second term describes the contribution of social norm imitation ($1-w$) to the change of $f_{s,n}$.

In the limit of weak selection ($\beta \to 0$), at the extremes where only one component of the type evolves (i.e., when $w = 0$ or $w = 1$), Eq. S16 is equivalent to the classical replicator dynamics[1], which we have employed throughout the rest of the paper. For this reason, in our simulations of this more general model we set $\beta = 0.01$, which reflects a weak selection regime[16].

2 Numerical methods

In general, an individual's behavior is characterized by three components:

- The strategy (s), defined in the section Interaction strategies.
- The social norm (d), defined in the section Social norms.
- The aggregation rule (q,M), defined in the section Aggregation of multiple assessments.

To study the dynamics of strategies (Strategy dynamics) and reputations (Reputation dynamics), we first define the strategy space S (i.e., the set of strategies competing against each other) and determine which components are allowed to change (i.e., evolvable) and which remain fixed. We then solve the reputation and replicator dynamics iteratively, as we describe below. Our iterative approach assumes that the timescale of reputation updates is faster than that of strategy updates, a common assumption in studies of indirect reciprocity.

2.1 Solving the reputation dynamics

To solve for the equilibrium of the reputation dynamics, we integrate Eq. S15 numerically. To do so, we initialize the system using the initial conditions for the strategy frequencies $f_i(t=0)$ and the reputations $r_{ij}(t=0)$ (i.e., each strategy's initial view of each strategy in the population). The strategy frequencies f_i subsequently remain fixed until the reputations reach an equilibrium.

Bistability in reputations. Prior work has assumed a unique average reputation equilibrium regardless of the initial condition $r(t=0)$ (i.e., the average reputation at the onset of the strategic updates). Here, we find that this is not always the case. Instead, in certain parameter regimes, bistability is possible: different average reputation equilibria will be reached depending on whether $r(0)$ is closer to 0 or to 1 (i.e., whether individuals start with an initially pessimistic versus initially optimistic view of the world).

For all our results, we assume that individuals start with an optimistic view—i.e. before the very first update of reputation is made, they start by regarding everyone as good $r(t=0)=1$—in keeping with the common assumption in studies of direct reciprocity that conditional strategies start by being 'nice'[17]. However, we also note that bistability only arises for $M>2$, and thus, the optimism assumption is unlikely to affect the robustness of 'look twice, forgive once'.

2.2 Solving the replicator dynamics

To solve for the equilibrium of the replicator dynamics, we integrate the replicator ODEs (Eq. S1) numerically. To do so, we first fix the strategy frequencies at their initial values $f_i(t=0)$ and solve for the equilibrium reputations $r^*_{ij}(t=1)$ as described above. Then, we fix the reputations at their equilibrium values $r^*_{ij}(t=1)$ and integrate a time-step of the replicator dynamics to obtain the new strategy frequencies $f_i(t=1)$. We then obtain new equilibrium reputations $r^*_{ij}(t=2)$ by integrating the reputation ODEs while keeping the strategy frequencies fixed at $f_i(t=1)$, after which we obtain new strategy frequencies $f_i(t=2)$ by integrating the replicator ODEs while keeping the reputations fixed at $r^*_{ij}(t=2)$. We repeat this procedure until the absolute value of the derivative reaches $\leq 10^{-10}$, at which point we consider the system to have reached an equilibrium.

2.3 Sampling initial frequencies

The method for sampling initial frequencies varied depending on the number of strategies considered in the strategy space.

For a strategy space with 4 or fewer strategies ($|S|\leq 4$), we varied the frequency of each strategy $f_i \in [0.05, 0.95]$ in increments of $\Delta f_i = 0.05$, excluding the tuples for which $\sum_{i\in S} f_i > 1$. This gives us 19 initial frequencies for 2 strategies, 171 for 3 strategies, and 975 for 4 strategies.

For a larger strategy space ($|S|>4$), we sampled initial frequencies uniformly at random from the $(|S|-1)$-simplex of all possible strategy frequencies. To obtain a sample of n initial frequencies, we applied the following algorithm: (1) take a random sample of $n-1$ numbers uniformly $\in (0,1)$ and sort them, (2) append 0 at the beginning and 1 at the end of the list to obtain $n+1$ numbers, and finally (3) take the differences between consecutive numbers in the list. This algorithm generates uniform samples from the simplex and is equivalent to sampling the $|S|$-dimensional Dirichlet distribution[18].

2.4 Numerical integration

Numerical integration was performed in the programming language `Julia v1.8.5`[19] using the package for Ordinary Differential Equations: `DifferentialEquations v7.7.0`[20]. The reputation ODEs were integrated using the `Tsit5` algorithm with a relative and absolute tolerance of $1e-8$ and an integration time

of $1e+3$. Every time step of the replicator ODEs was integrated using the `RadauIIA3` algorithm, with an absolute tolerance of $1e-15$. After every step, we checked the absolute magnitude of the derivative to set the integration time for the next step. We stopped integration when the absolute size of the derivative was $< 1e-12$.

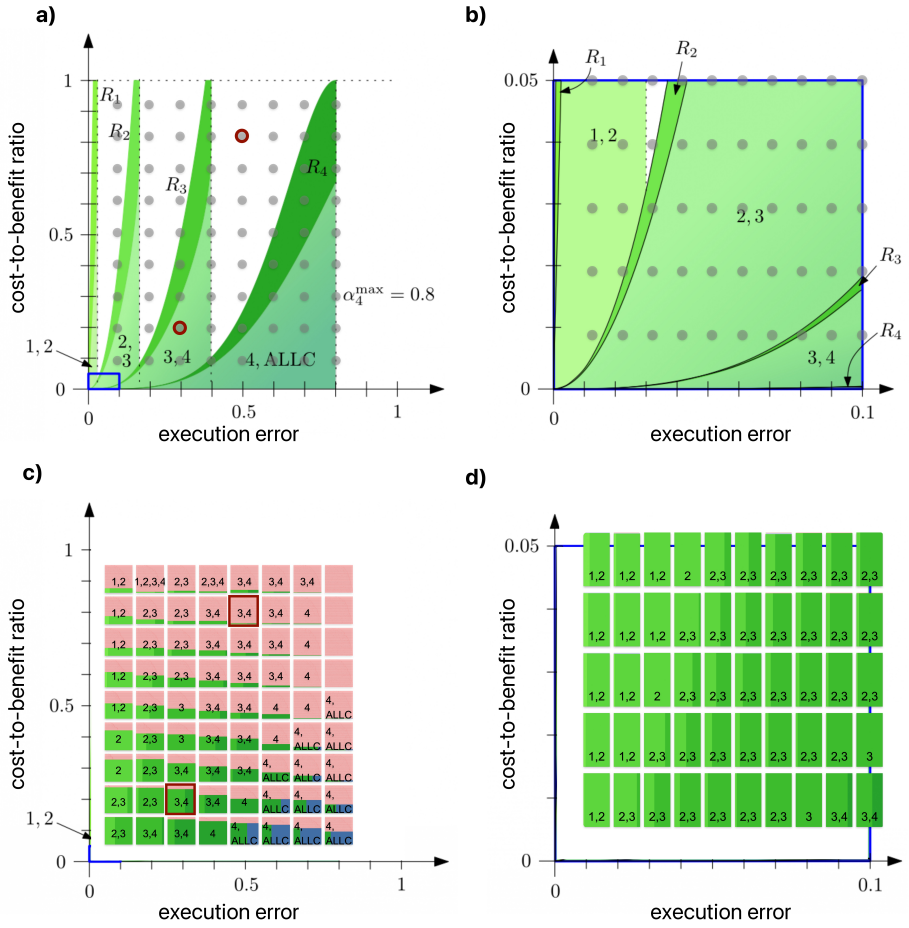
Simplexes with evolutionary dynamics. For the 2-simplexes shown in Fig. 1 of the main text, we fixed the social norm for the whole population, selected the fixed parameters q and M to characterize the specific type of $\mathrm{DISC}_{q,M}$, and allowed the 3 strategies to compete.

To determine the equilibria and their stability, we first solved for the equilibrium of the replicator ODEs under fixed values of the error rates and the benefit and cost of cooperation. Then, we solved the equilibrium reputations at each frequency equilibrium and selected the ones with feasible values (i.e., real solutions of the reputation dynamics between 0 and 1 that, when substituted in the frequency equilibria, also yield real numbers between 0 and 1). Lastly, we performed a linear stability analysis at each of the feasible equilibria and reported the stable equilibria as solid disks and the unstable equilibria as empty disks.

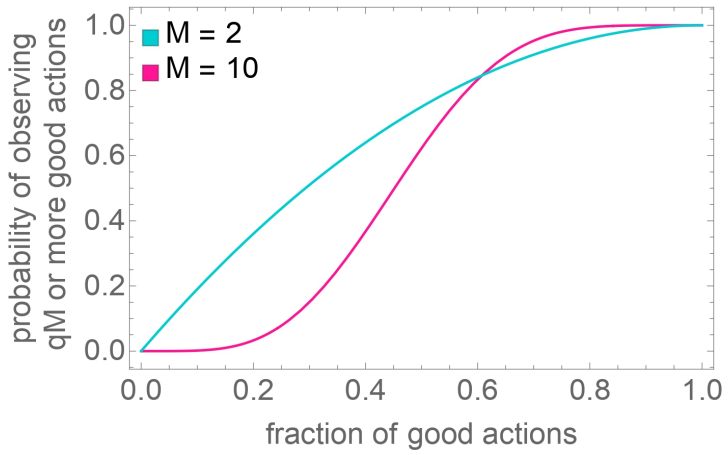
To plot the gradients of selection, we first solved for the equilibrium reputations at different initial frequencies in the interior of the simplex. Then, we substituted the equilibrium reputations into the replicator equations to obtain the directional field of the dynamics. Lastly, we used the function `ListStreamPlot` of the `Mathematica`[21] software to interpolate and plot the arrows.

Simplexes with temporal trajectories. For the 2-simplexes in Extended Data Fig. 1 and the faces of the 3-simplex in Fig. 2, we fixed the social norm and reported the temporal trajectories starting from the interior of the simplex and ending at a stable equilibrium. The initial frequencies were selected deterministically as described above (see Sampling initial frequencies), where each strategy frequency f_i took values $f_i \in \{0.05, 0.1, ..., 0.85, 0.9\}$, yielding a total of 171 different initial frequencies. Then, we integrated numerically as described above (see Numerical integration) but using the `RadauIIA5` algorithm[20] and saving every time step until convergence. We reported the steady states that the temporal dynamics reached as stable (filled disks) and the vertices of the simplex that were not reached by the trajectories as unstable (empty disks). Finally, we plotted the full temporal trajectories in the plane and transformed the coordinates to the simplex using the function `GeometricTransformation` of the `Mathematica`[21] software.

3 Supplementary Figures



Supplementary Figure S1: Framework comparison with Berger & Grune [22]. Panels (**a**) and (**b**) are reprinted from Fig. 3 of *Games and Economic Behavior*, Vol 98, Ulrich Berger and Ansgar Grüne, "On the stability of cooperation under indirect reciprocity with first-order information", Pages No.19-33, Copyright (2016), with permission from Elsevier. Such panels show the regions of the (cost-to-benefit ratio, execution error)-parameter space where either pure (dark green) or mixtures (light green) of aggregating discriminators are ESS. Numbers indicate the value of $(1-q)M$, e.g., 3 corresponds to $DISC_{2/5,5}$, per the original notation in Berger & Grune [22]. Panel (b) is a zoom-in of the small blue rectangle anchored at the origin in panel (a). We added the gray disks to mark the combinations of parameters that we sampled to investigate the outcome of replicator dynamics and to estimate the basins of attraction towards distinct equilibria. In (c) and (d), each small bar corresponds to a disk in (a) and (b), respectively, and summarizes the size of the basins of attraction for the steady states reached by numerical integration of 100 different initial frequencies. Basins of attraction of the discriminator-only equilibria are shown in shades of green (the darker the green, the more tolerant the discriminator strategy). Basins of attraction of equilibria where discriminators coexist with *ALLC* are shown in green (corresponding to the discriminator) and blue (corresponding to *ALLC*). Basins of attraction where discriminators are neutral with *ALLD* (and, thus, no cooperation is sustained) are shown in pink. Maroon outlines around gray disks in (a) correspond to maroon outlines around bars in (c).



Supplementary Figure S2: Probability of sampling good actions for different numbers of observations. Curves show the probability that an individual observes qM or more good actions, as a function of the fraction of all good actions that a donor performed in the last round. A population made up solely of aggregating discriminators resides in the right-most section of this plot, where the magenta curve exceeds the teal curve.

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